

CPE326 Operating System Final 2022

Part2-1

1. What is the benefit of thread over process when context switching occurs?
2. Explain how can a race condition effect the result of concurrency execution?
3. Give an example of code to create a lock from test_and_set()
4. How can you eliminate the time wasted in spinning loop in lock?

↳ yield

การที่ process Interrupt

main thread จะไปรอจนกว่า global variable จะว่าง

while (testandset(flag → lock) == 1)
//

1.) Thread switch will keep the virtual memory space remains the same.

2.) Race condition happens when 2 or more processes access the resource at the same time

Ex.	Thread 1	Thread 2
	load counter	
	add 1	
	store counter	load counter
		add 1
		store counter

3.) void critical() {
while (testandset(&lock) == 1);
critical section
lock = 0;
}

4.) Yield CPU and do other process.
do sleep

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CPE 326 Operating System Final Exam

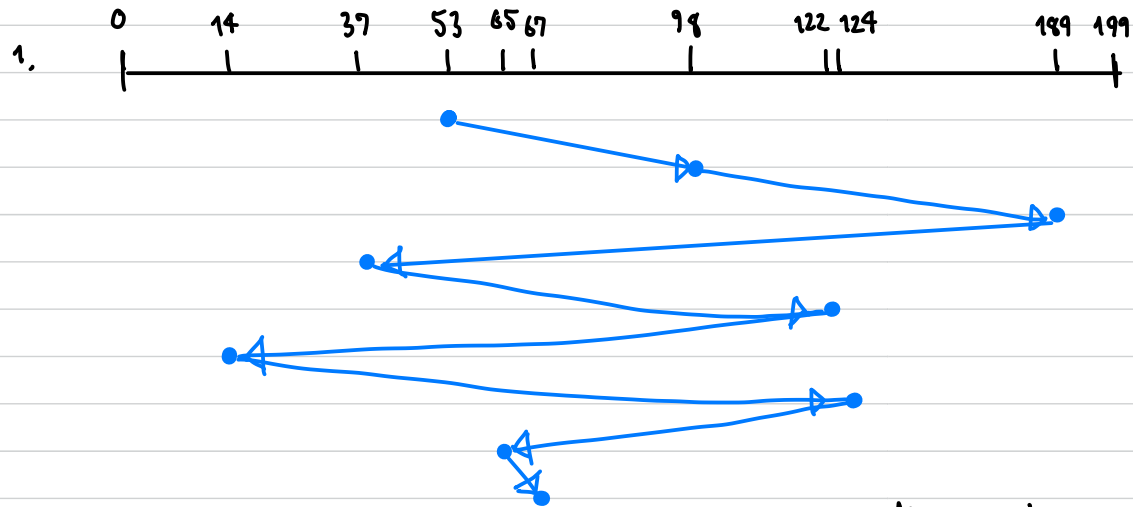
Part 2-2

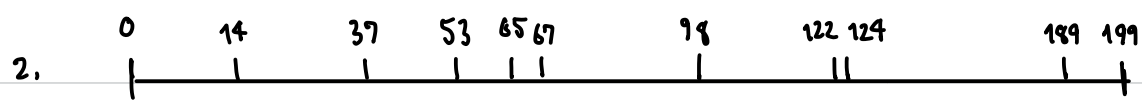
How can we prevent deadlock in dining philosopher problem? Explain in detail.

We use semaphore to count the available chopsticks for philosophers, we can call this solution "the waiter solution". The waiter always share chopsticks until there is only one chopstick remain unused. We assume that all philosophers must request their left chopstick first then their right. If a right chopstick is requested, request for left chopstick will be deferred.

If a disk has cylinder 0-199, and a head starts at position 53. The following positions are in reading queue {98,189,37,122,14,124,65,67}. Illustrate the seek path and calculate the total head movement for the following algorithms:

1. FCFS
2. SSTF
3. C-SCAN (left to right)
4. C-LOOK (left to right)

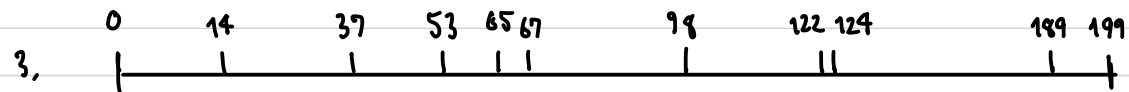




สิริวิชัย วัฒนกุล

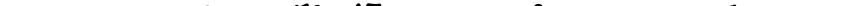
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Movement = 242



C-Scan su Min-Max

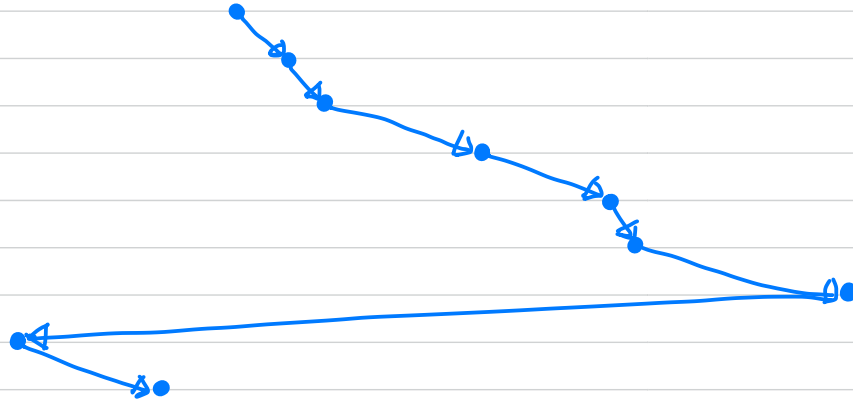
Movement = 382

4. 

C-Look ในกรณี Min-Max

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Movement = 334